$a_n =$

$$a_n = \begin{cases} \dfrac{1}{(n-4)(n-7)} & \text{if } n \neq 7, n \neq 4 \\ 10 & \text{if } n = 4 \\ 2 & \text{if } n = 7 \end{cases}$$

At $n = 4$, $a_4 = 10$

At $n = 7$, $a_7 = 2$

But for large n, for example $n = 10$

$$a_{10} = \frac{1}{(10-4)(10-7)} = \frac{1}{6 \cdot 3} = \frac{1}{18}$$

At $n = 100$,

$$a_{100} = \frac{1}{96 \cdot 93}$$ which is extremely close to 0.

So, although two terms are specially defined:

$a_4 = 10$, $a_7 = 2$

The terms eventually become very close to 0.

Therefore, the sequence converges to zero.

If you change only a finite number of terms of a convergent sequence, its limit doesnot change.

It means: If you change only a finite number of terms of a convergent sequence, its limit doesnot change.

For exa: consider $\frac{1}{n}$

We know, $\frac{1}{n} \to 0$

Now suppose I change few terms:

$a_1 = 100$

$a_2 = -500$

$a_{50} = 1000$

but leave every other term equal to 1/n.

Example-2

$a_n = \frac{1}{n-4}$ if $n \neq 4$ Is it a sequence?

let $n = 1, 2, 3, 4$

$a_1 = \frac{1}{1-4} = \frac{-1}{3}$

$a_2 = \frac{1}{2-4} = \frac{-1}{2}$

$a_3 = \frac{1}{3-4} = -1$

But when we reach $n = 4$

$a_4 = \frac{1}{4-4} = \frac{1}{0}$

And $\frac{1}{0}$ is undefined

So there is no value for a_4.

The sequence would look like: $\frac{-1}{3}, \frac{-1}{2}, -1, ?, 1, \frac{1}{2}, \frac{1}{3} \cdots$

There is a missing term at $n=4$.

So is it a sequence?

If the definition is ONLY

$$a_n = \frac{1}{n-4}, \quad n \neq 4$$

Then, No, it is not a sequence indexed by all $n \in \mathbb{N}$.

Because a_4 does not exist.

If we seperately give a value for a_4, then we can make it a proper sequence.

$$a_n = \begin{cases} \frac{1}{n-4}, & n \neq 4 \\ 10, & n = 4 \end{cases}$$

Now every n has a value::

$a_1 = -\frac{1}{3}$

$a_2 = \frac{-1}{2}$

$a_3 = -1$

$a_4 = 10$ etc

Now there is no missing term.

So, this is a sequence.

$\{a_1, a_2, a_3, a_n, a_{n+}, \}$ is some finite elements are elimated will it still converge.

$a_n \to a \Rightarrow \forall \varepsilon' > 0, \exists N_1$ such that $\forall n \geq N_1$, $a_n \in (a-\varepsilon', a+\varepsilon')$

The definition of convergence says:

$\forall \varepsilon' > 0, \exists N_1$

such that whenever $n \geq N_1$

we have, $|a_n - a| < \varepsilon'$

The interval version of this is:

$a_n \in (a-\varepsilon', a+\varepsilon')$

Since, $a_n \to a$. I can make a_n as close to a as I want by going far enough in the sequence.

$\{b_n\} \to b \Rightarrow \forall \varepsilon'' > 0, \exists N_2$ such that for every $n \geq N_2$ we have,

$b_n \in (b-\varepsilon'', b+\varepsilon'')$

In absolute value language,

$|b_n - b| < \varepsilon''$

Q?:

$\forall \varepsilon = \varepsilon' + \varepsilon'' > 0, \exists = ?$

① $N = \max\{N_1, N_2\}$ ✓ Suppose $N_1 = 10$ and $N_2 = 20$. If we choose $N = 20$ then $n \geq 20$ automatically means $n \geq 10$ and $n \geq 20$. So Both conditions work.

② $\min\{N_1, N_2\}$ ✗ Suppose $N_1 = 10$, $N_2 = 20$ If you choose $N = \min(10, 20) = 10$, then $n \geq 10$ does not guarantee $n \geq 20$ so the second condition might not work.

③ $N_1 + N_2$

④ $N_1 \times N_2$

Example:

$b_n = \frac{3n+1}{2n^2+n}$, why does it converge? show:

Step 1: $b_n = \frac{3n+1}{2n^2+n}$

The denominator can be factored:

$2n^2 + n = n(2n+1)$

$\frac{1}{n} + \frac{1}{2n+1}$

Take common denominator:

$\frac{2n+1}{n(2n+1)} + \frac{n}{n(2n+1)}$

Add: $= \frac{2n+1+n}{n(2n+1)}$

And because: $n(2n+1) = 2n^2+n$

we get. $\frac{3n+1}{2n^2+n} = \frac{1}{n} + \frac{1}{2n+1}$

Prove $-n^2$ converge or not

let $a_1 = -1$, $a_2 = -4$, $a_3 = -9$

So, $-1, -4, -9, -16, -25 \ldots\ldots$

They are not approaching 0.

$-n^2 \to -\infty$

So it doesnot converge to 0.

Prove that $\frac{1}{(-2)^n}$ converge to 0

we need to show that for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n > N \Rightarrow |a_n - 0| < \varepsilon$.

Find $|a_n - 0|$

$$|a_n - 0| = \left|\frac{1}{(-2)^n}\right|$$

Since, $|(-2)^n| = 2^n$

we get $|a_n - 0| = \frac{1}{2^n}$

So our goal is:

$$\frac{1}{2^n} < \varepsilon$$

Taking reciprocals gives

$$2^n > \frac{1}{\varepsilon}$$

Take $\log_2$,

$$n > \log_2\left(\frac{1}{\varepsilon}\right)$$

So we can choose N such that

$$N > \log_2\left(\frac{1}{\varepsilon}\right)$$

let $\varepsilon > 0$

Choose N such that $N > \log_2\left(\frac{1}{\varepsilon}\right)$

If $n > N$, then $n > \log_2\left(\frac{1}{\varepsilon}\right)$

Therefore,

$$2^n > \frac{1}{\varepsilon}$$

$$\frac{1}{2^n} < \varepsilon$$

But, $\frac{1}{2^n} = \left|\frac{1}{(-2)^n} - 0\right|$

Hence, $\left|\frac{1}{(-2)^n} - 0\right| < \varepsilon$

$\therefore$ By defn of convergence, $\frac{1}{(-2)^n} \to 0$